ISE 218: Fundamentals of Information Technology

Stony Brook Institute at Anhui University

Homework #3

Spring 2025

Assignment Due: March 28, 2025

Neatly write or type your answers to the following problems on a separate sheet of paper and submit a digital copy. For mathematical calculations you must show all work to receive full credit and simplify your answers as much as possible, giving a single number as your answer.

For all questions involving sizes, you should use 2-based units of size. For example, 1 KB = 2^{10} bytes, not 10^3 bytes.

1. [6 pts] How many bits are required to address a 4K × 16 main memory if:
   a. main memory is byte-addressable?
      4K × 16 memory means there are 4096 (2^{12}) words, each word is 16 bits (2 bytes).
      If byte-addressable, each byte requires a unique address.
      Total bytes = 4096 × 2 = 8192 = 2^{13}.
      Bits required: 13 bits

   b. main memory is word-addressable?
      4K × 16 memory means there are 4096 (2^{12}) words, so
      Bits required: 12 bits

2. [9 pts] Suppose that a 4M × 64 main memory is built using 512K × 16 RAM chips and memory is word addressable.
   a. How many RAM chips are necessary?

      4M × 64 memory: 4M words, each word is 64 bits.
      Each RAM chip: 512K × 16 bits.
      Number of chips needed = (4M × 64) / (512K × 16)
      = $(4 \times 2^{20} \times 64) / (512 \times 2^{10} \times 16)$
      = $(4 \times 2^{10} \times 64) / (512 \times 16)$
      = (256) / (8)
      = 32 chips

b. How many address bits are needed to address one byte in a RAM chip?

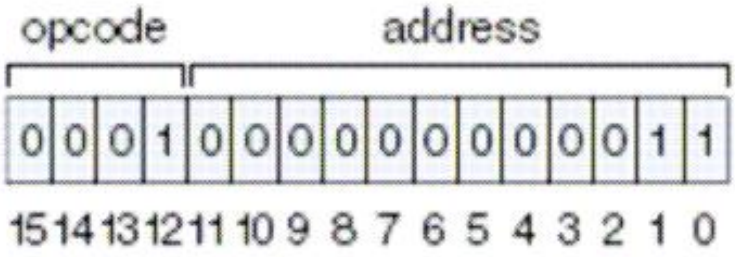
Each RAM chip is 512K × 16 bits.
512K = 2^{19} bytes.
Each word has 2 bytes, so we need an additional bit to address the individual bytes within a word.
Address bits required: 19 + 1 = 20bits

c. How many address bits are needed for all memory?

Total memory: 4M × 64 bits.
4M = 2^{22} words.
If word-addressable, address bits required: 22bits
If byte-addressable, each word has 8 bytes, so we need 3 additional bits to address the individual bytes within a word.
Address bits required: 22 + 3 = 25bits

3. [3] Write 3 MARIE Instruction set for Load, Jump, and Halt in binary.
Load: Opcode 0001 (4 bits) + 12-bit address. Answer: 0001 0000 0000 0000.
Jump: Opcode 1001 (4 bits) + 12-bit address. Answer: 1001 0000 0000 0000.
Halt: Opcode 0111 (4 bits) + 12 zeros. Answer: 0111 0000 0000 0000.

Instruction Number		Instruction	Meaning
Bin	Hex		
0001	1	Load *X*	Load the contents of address *X* into AC.
0010	2	Store *X*	Store the contents of AC at address *X*.
0011	3	Add *X*	Add the contents of address *X* to AC and store the result in AC.
0100	4	Subt *X*	Subtract the contents of address *X* from AC and store the result in AC.
0101	5	Input	Input a value from the keyboard into AC.
0110	6	Output	Output the value in AC to the display.
0111	7	Halt	Terminate the program.
1000	8	Skipcond	Skip the next instruction on condition.
1001	9	Jump *X*	Load the value of *X* into PC.



4. [6] Write 1 sets of micro-operations using register transfer language (RTL) for ADD.
   MAR ← X
   MBR ← M[MAR]
   AC ← AC + MBR

5. [7] Explain SKIPCOND (400), and how SKIPCOND is related with the AC register.
   SKIPCOND(400):
   Condition code 400 in hex = 0100 0000 0000 in binary.
   The first two bits (01) indicate the condition: skip if AC = 0.
   Relation to AC:
   SKIPCOND directly checks the value of the Accumulator (AC).
   If AC satisfies the condition (e.g., AC = 0), the next instruction is skipped.

6. [5] In MARIE Architecture how 'fetch, decode, get operand and execute' works for Sub operation.
   Fetch:
   MAR ← PC
   IR ← M[MAR]
   PC ← PC + 1
   Decode:
   MAR ← IR[11:0]
   (Decode IR[15:12])
   Get operand:
   MBR ← M[MAR]
   Execute:
   AC ← AC - MBR

7. [4] What is the difference between AC and MBR?

   AC (Accumulator):
   Primary register for arithmetic/logic operations.
   Holds intermediate results and final outputs.
   MBR (Memory Buffer Register):
   Temporarily holds data read from or written to memory.
   Acts as a buffer between CPU and memory.
   Key Difference:
   AC is directly used in computations, while MBR is a temporary storage for memory-CPU transfers.